AQUEN FLUID TECHNOLOGIES LLP

Complete Water & Wastewater Treatment Solutions

Design | Manufacturing | Supply | Installation | Commissioning | AMC

ULTRAFILTRATION (UF) SYSTEM

TECHNICAL CATALOGUE

CAPACITY RANGE : 2 m³/Hr TO 20 m³/Hr | OPERATION HOURS : 16 HRS

- High Filtration Efficiency (0.01 – 0.1 µm)
- Removes Suspended Solids, Bacteria & Viruses
- Compact & Modular Design
- Automatic Operation with PLC
- Low Chemical Consumption
- Easy Operation & Maintenance
- Suitable for Water & Wastewater Applications
- Skid Mounted / Containerized Options
- Customized Design as per Requirement

CLEANER WATER BRIGHTER TOMORROW



UF PLANT – 5 m³/Hr (Installation in Progress)



UF PLANT – 10 m³/Hr (Complete Installation)



UF PLANT – 15 m³/Hr (Industrial Application)



UF PLANT – 20 m³/Hr (Skid Mounted)

PROCESS FLOW DIAGRAM – UF SYSTEM



UF MEMBRANE MODULE



PVDF Hollow Fiber UF Membranes

- Pore Size : 0.01 – 0.1 µm
- High Strength PVDF Material
- High Flux & Long Life
- Backwash & Chemical Cleaning
- Removes TSS, Bacteria, Viruses, Colloids & Turbidity

KEY FEATURES

- Pore size 0.01 – 0.1 µm
- High quality filtrate water
- Automatic backwash & CIP
- PLC based control panel
- Low power consumption
- Compact skid design
- Wide range 2 – 20 m³/Hr
- Suitable for various industries
- SS / MS Skid options
- Easy installation & operation

APPLICATIONS / INDUSTRIES

Residential Societies | Commercial Buildings | Hospitals & Healthcare | Hotels & Resorts

Food & Beverage Industry | Pharmaceutical Industry | Textile Industry | Power Plants

Industrial Process Water | Municipal Water Treatment | RO Feed Pre-Treatment | Wastewater Reuse & Recycling

TECHNICAL SPECIFICATIONS – UF SYSTEM (OPERATION HOURS : 16 HRS)

Parameter	Unit	2 m³/Hr	5 m³/Hr	8 m³/Hr	10 m³/Hr	15 m³/Hr	20 m³/Hr
Design Flow	m³/Hr	2	5	8	10	15	20
Operation Hours	Hrs	16	5	8	16	16	16
Total Daily Flow	m³/day	32	80	128	160	240	320
Recovery (Filtrate)	%	92	92	92	92	92	92
Filtrate Flow	m³/Hr	1.84	4.6	7.36	9.2	13.8	18.4
Reject Flow	m³/Hr	0.16	0.4	0.64	0.8	1.2	1.6
Membrane Type	-	PVDF Hollow Fiber (Outside-In)					
Pore Size	µm	0.01 – 0.1					
Design Flux	LMH	60 – 80					
No. of Modules (Typical)	Nos.	1	2	3	4	6	8
Membrane Area (Approx.)	m²	60	120	180	240	360	480
Feed Pump Capacity	m³/Hr @ m	2.5 @ 20	6 @ 20	8.5 @ 20	11 @ 20	16 @ 20	22 @ 20
Backwash Pump	m³/Hr @ m	5 @ 20	8 @ 20	12 @ 20	15 @ 20	20 @ 20	25 @ 20
CIP Pump	m³/Hr @ m	3 @ 20	5 @ 20	5 @ 20	5 @ 20	8 @ 20	8 @ 20
Connected Load (Approx.)	kW	2.5	4.0	5.5	7.0	10.0	13.0
Required Area (Approx.)	m²	8 – 10	10 – 15	15 – 20	18 – 25	25 – 35	35 – 50

TYPICAL INLET & OUTLET PARAMETERS

Parameter	Unit	Inlet (Typical)	Outlet (UF Filtrate)
Turbidity	NTU	5 – 50	< 0.1
TSS	mg/L	10 – 100	< 1
COD	mg/L	< 100	< 10
BOD	mg/L	< 50	< 5
Oil & Grease	mg/L	< 10	< 1
SDI (15 min)	-	< 5	< 2
pH	-	6.0 – 9.0	6.0 – 9.0
Total Coliform	MPN/100 mL	$10^3 – 10^4$	< 10^2

Note: Above values are typical and can be customized as per project requirement.

MOC (MATERIAL OF CONSTRUCTION)

Item	Material
UF Membrane Housing	UPVC / uPVC PVDF /SS-304
Process Piping (Feed/Filtrate/Backwash)	UPVC SCH-80 (Standard) CPVC / SS-304 (Optional)
Valves	UPVC / CPVC / SS-304
Pumps (Feed/Backwash/CIP)	CI / SS-304 (as per duty)
Skid	MS Epoxy Painted / SS-304 (Optional)
Cartridge Filter Housing	UPVC / SS-304
Instrumentation	Pressure Gauge, Flow Meter, Turbidity Meter, Level Switches, etc.
Control Panel	MCC with PLC (Standard)

TYPICAL BILL OF MATERIAL (BOM) – 10 m³/Hr UF PLANT

Sr. No.	Description	Specification	Qty.	Sr. No.	Description	Specification	Qty.
1	Feed Pump	11 m³/Hr @ 20 m, SS-304	1 No.	6	CIP Pump	5 m³/Hr @ 20 m	1 No.
2	Cartridge Filter Housing	5 µm, UPVC / SS-304	1 No.	7	Piping & Valves	UPVC SCH-80	1 Set
3	UF Membrane Modules	PVDF, 60 m² / module	4 Nos.	8	Instrumentation	PG, FM, Turbidity Meter, etc.	1 Set
4	Membrane Housing	UPVC	4 Nos.	9	Skid	MS Epoxy / SS-304	1 Set
5	Backwash Pump	15 m³/Hr @ 20 m	1 No.	10	Control Panel	MCC with PLC	1 Set

OPTIONAL ITEMS

- MGF & ACF (Pre-Treatment)
- Chemical Dosing System (CIP)
- CIP Tank with Heater (Optional)
- Treated Water Storage Tank (TTP for Flushing)
- Online Monitoring (pH, Turbidity)
- SS-304 Skid & Piping
- Containerized Package

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1. UF - 10 m³/hr (Civil Work in Progress)
MGF + ACF + UF System



2. UF - 10 m³/hr (Installation in Progress)
MGF + ACF + UF + Dosing + Control Panel



3. UF - 15 m³/hr (Fully Installed)
MGF + ACF + UF + CIP + Dosing + Automation



4. UF - 20 m³/hr (MSRL Tank + E&M Installation)
MGF + ACF + UF + Treated Water MSRL Tank



5. UF - 5 m³/hr (MSEP Tanks + Installation)
MGF + ACF + UF System



6. UF - 10 m³/hr (Open Installation – Complete Plant)
MGF + ACF + UF + Treated Water Tank + Piping

